

Grid-connected control of PV-Wind hybrid energy system

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Article Info

Article history:

Received Oct 14, 2020

Revised Mar 22, 2021

Accepted Apr 23, 2021

Keywords:

Grid

Hybrid energy system

MPPT

OTSR

Photovoltaic

PMSM

Wind

ABSTRACT

This paper presented a strategy for modeling, simulation and control of a hybrid grid connected power system which is in fact a rather complex system. In this work, we study how to use two renewable energies in an efficient manner without any disturbing of the main network. Our hybrid energy system (HES) is composed by two renewable energy sources, the photovoltaic source and wind source. It is better for these two systems to work at their maximum power in order to return the investment cost of the system. The proposed solution is to connect those generators to the electrical grid via the AC bus with trackers. Where the photovoltaic generator (PVG) is followed by a DC/DC boost converter, controlled by a perturb and observe (P&O) tracker, then followed by a three-phase voltage inverter (3-ph-VSI) which is controlled by the watt-var decoupled method. The wind system is based on permanent magnet synchronous machine (PMSM) which is used as a variable speed generator and directly connected to the turbine (without gearbox) followed by back-to-back converters. The grid side converter (GSC) allows us to control the DC bus voltage and unity power factor, while the machine side converter (MSC) ensures us to control the PMSM speed. Knowing that pitch angle control is not considered in our study. The MPPT control strategy, which is based on the optimum peak speed ratio (OTSR), is used to ensure the most energy efficiency despite variations in wind speed. The coupling of the two systems is done via the Point of Common Collecting (PCC). Finally, simulation results show the feasibility of our solution and the good performances of the proposed control strategy, applied for medium-sized HES in power generation. They also demonstrate that such system topology is very advantageous.

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1. INTRODUCTION

Along with the crisis of the exhaustion of the traditional energy resources, and with environmental damage caused by increased carbon dioxide emissions, the use of the renewable energy has become more and more one of the faster growing and clean energy resources [1]-[3]. Wind and photovoltaic energy sources can be considered as the most promising technologies for generating electricity [4]. However, solar irradiation and wind are intermittent and highly dependent on climatic conditions [5].

Hybrid energy system (HES) combines renewable and conventional energy sources. However, for the applications of autonomous SEHs, storage is essential when the grid is missing or when it is necessary to ensure continuity of service [3]-[6] which becomes an economic constraint. However, a possible connection

to the grid would lead to minimize the requirement of the storage system [7], [8]. HES is advantageous in view of the reduction of the overall cost of the system, the response to the demand of the AC loads [9] and also the demand of the load of the network is relieved. HES systems cause problems when it is connected to the network. The first is maximise the power delivered by controllers depending of climatic or load variations [10], and synchronization with the power grid [11]. The proposed solution is to connect HES to the network via the AC bus with trackers.

Our studied HES system is characterized by two energy sources without a storage system in order to minimize the cost of the system. These two energy sources will be coupled to the electricity network via the AC bus. As shown in Figure 1, the photovoltaic generator (PVG) is followed by a DC/DC converter; the latter will be controlled by the Perturb and Observe (P&O) algorithm, it is a technique to control MPPT (Maximum Power Point Tracking) where many researchers have focus on the implementation of this method [12], followed by a three-phase inverter (3-ph VSI) controlled by the decoupled Whatt-Var approach to be able to control the active and reactive powers that we need to inject to the grid. Two back-to-back converters power are used for the wind system energy. It is composed of a PMSG, its control strategy is used for the GSC to realize the decoupled current control concept to control active and reactive power and to maintain a constant DC bus voltage [13], [14]. The Field Oriented Control (FOC) is used by the MSC to control the speed of the PMSG [15].

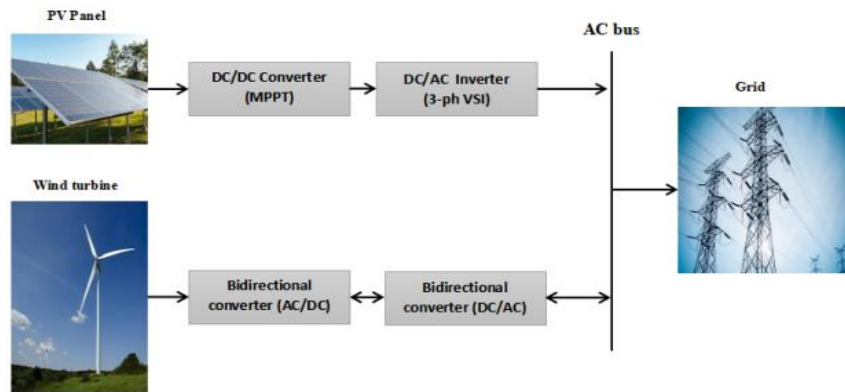


Figure 1. Proposed configuration of grid connected HES (PV/Wind)

We subdivided our work into three parts, where we will present the analysis, modeling and control of the grid connected PV system, the grid connected wind system and the grid connected PV/Wind system. we finalize it by giving the result of simulation of our system under MATLAB/Simulink, a discussion of the results obtained and a conclusion.

2. GRID CONNECTED PV/ WIND SYSTEM

2.1. Grid conected PV system

Grid connected PV system studied shown in figure 2 is composed by photovoltaic modules (KC200GT modules which is tested and well suited to traditional photovoltaic applications [16]), MPPT controller, boost converter, three-phase Inverter (We choose those without transformer due to its high efficiency and low cost [17]), output self-filter and finally a circuit breaker to connect our system to the power grid [18].

The PV generator generates a DC voltage, so it is necessary to use an inverter to convert DC voltage into an alternative voltage (AC) [9]. In our study, a three-phase bidirectional DC/AC inverter was done by the decoupled Whatt-Var approach. This approach purpose is to control the active power (P) and the reactive power (Q). At Common Coupling Point (CCP), we measured the three-phase Voltages (V_{abc}) [19], [20]. From the voltage control loop, witch is considered to control the voltage of the DC side [8], we have :

$$\begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} = \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} - L \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (1)$$

Applying Park's transformation method, (1) can be rewritten as follow [18], [21], [22]:

$$\begin{cases} u_d = v_d - L \frac{di_d}{dt} + \omega_0 \cdot L \cdot i_q \\ u_q = v_q - L \frac{di_q}{dt} - \omega_0 \cdot L \cdot i_d \end{cases} \quad (2)$$

The d-q control is considered for transformation of $V_{a,b,c}$ known as grid voltages. It can be deduced:

$$\begin{cases} u_d = v_d - K_p \left(1 + \frac{K_i}{s}\right) (i_{dref} - i_d) + \omega_0 L i_q \\ u_q = v_q - K_p \left(1 + \frac{K_i}{s}\right) (i_{qref} - i_q) - \omega_0 L i_d \end{cases} \quad (3)$$

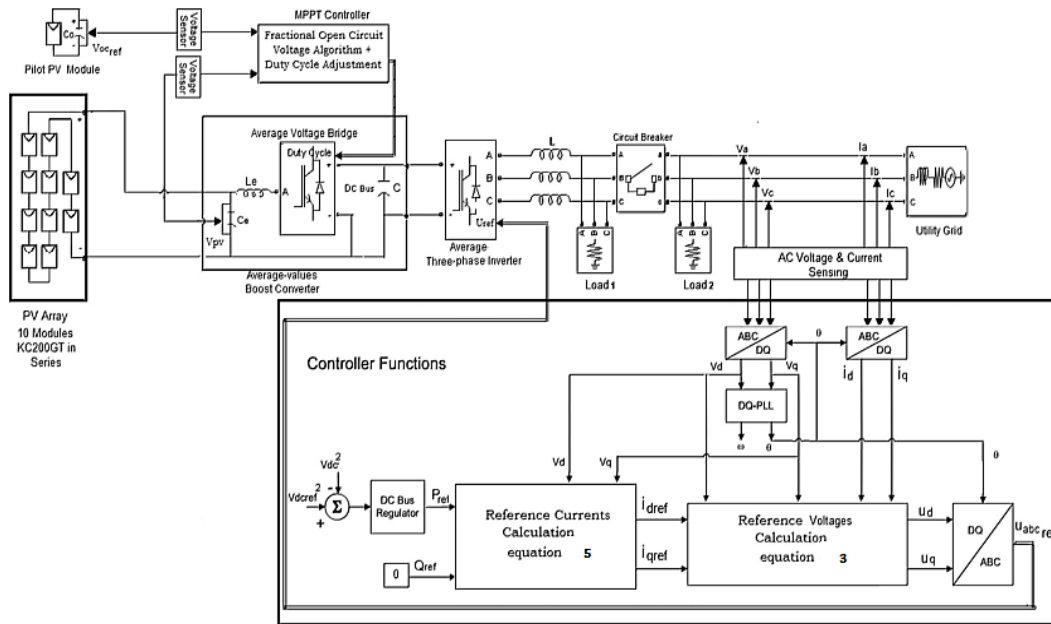


Figure 2. Main diagram of the PV system connected to the main Grid

Errors in system parameters must be compensated by the PI regulator. To control active and reactive power, we use the decoupled Watt-Var control [18], [23].

The active and reactive powers are expressed in (4) [18, 24]:

$$\begin{cases} P = \frac{3}{2} * (v_d * i_d + v_q * i_q) \\ Q = \frac{3}{2} * (v_q * i_d - v_d * i_q) \end{cases} \quad (4)$$

The currents references (i_{dref} , i_{qref}) can be expressed by (5):

$$\begin{bmatrix} i_{qref} \\ i_{dref} \end{bmatrix} = \frac{2}{3(v_d^2 + v_q^2)} \begin{bmatrix} v_d & v_q \\ v_q & -v_d \end{bmatrix} \begin{bmatrix} P_{ref} \\ Q_{ref} \end{bmatrix} \quad (5)$$

This decoupled Watt-Var control strategy allows the inverter to supply all the power generated by the photovoltaic module to the network and it also serves to control the DC bus voltage [11].

2.2. Grid connected wind system

The study wind energy conversion system is connected to the grid. The PMSG provides the power to the grid through back-to-back power converters and filter as illustrated in Figure 4 [25]

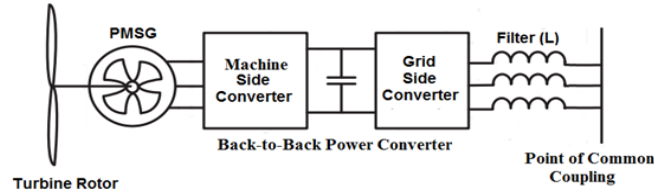


Figure 4. PMSG based wind energy conversion system [25]

2.2.1. Wind turbine modeling

The modeling of wind turbine helps to understand the dynamic and static behaviour of the wind system. The power output of wind turbine $P_t(W)$ is given by [26]:

$$P_t = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) V_v^3 \quad (6)$$

The tip speed ratio (λ) and the power coefficient (C_p) can be expressed as follows (7)-(9):

$$\lambda = \frac{\omega_m R}{v} \quad (7)$$

$$C_p(\lambda) = 0.5176 \left(\frac{116}{\lambda_i} - 5 \right) \cdot \exp^{\frac{21}{\lambda_i}} + 0.006795 \lambda_i \quad (8)$$

$$\lambda_i = \frac{1}{\frac{1}{\lambda} - 0.035} \quad (9)$$

Further, the Mechanical torque (T_t) presented in equation 12, can be deduced from (6):

$$T_t = \frac{P_t}{\omega_m} = \frac{1}{2\lambda} \cdot \rho \cdot \pi \cdot R^3 \cdot v^2 \cdot C_p(\lambda) \quad (10)$$

Aerodynamic torque coefficient ($C_t(\lambda)$) is determined by (11):

$$C_t(\lambda) = \frac{C_p(\lambda)}{\lambda} \quad (11)$$

Form equations (10) and (11), (T_t) can be rewritten as follow (12):

$$T_t = 0.5 \cdot \rho \cdot \pi \cdot R^3 \cdot v^2 \cdot C_t(\lambda) \quad (12)$$

Figure 5 and Figure 6 illustrated respectively the Wind turbine power-speed and $C_t(\lambda)$ characteristics for the study wind turbine.

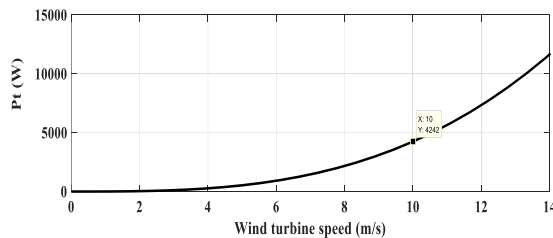
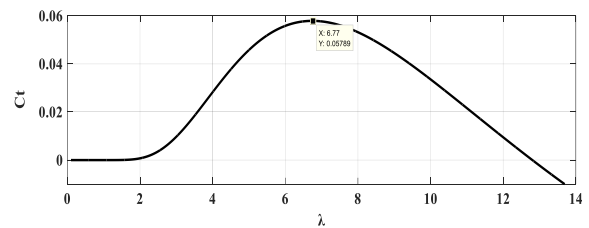


Figure 5. Wind turbine power-speed characteristic

Figure 6. $C_t(\lambda)$ characteristic

2.2.2. Modeling of the permanent magnet synchronous generator (PMSG)

For a simpler and easier analysis: The stator windings are connected in Y; losses by hysteresis and eddy current are neglected and no dynamic response process of the excitation current [27]:

Electrical equation of PMSG is expressed below:

$$v_{abc_s} = \frac{d\psi_{abc_s}}{dt} + R_s i_{abc_s} \quad (13)$$

Applying Park transform equation (13) becomes:

$$\begin{bmatrix} v_{d_s} \\ v_{q_s} \end{bmatrix} = [P(\theta_m)][R_s][P(\theta_m)]^{-1} i_{d_s} + [P(\theta_m)][P(\theta_m)]^{-1} \frac{d}{dt} [\psi_{d_s}] + [P(\theta_m)] \left(\frac{d}{dt} [P(\theta_m)]^{-1} \right) [\psi_{d_s}] \quad (14)$$

Where $P(\theta_m)$ is park transformation expressed as follow:

$$P(\theta_m) = \frac{2}{3} \begin{bmatrix} \cos(\theta_m) \cos(\theta_m - \frac{2\pi}{3}) \cos(\theta_m + \frac{2\pi}{3}) \\ -\sin(\theta_m) -\sin(\theta_m - \frac{2\pi}{3}) -\sin(\theta_m + \frac{2\pi}{3}) \\ \frac{1}{2} \frac{1}{2} \frac{1}{2} \end{bmatrix} \quad (15)$$

$$[\psi_{d_s}] = \begin{bmatrix} \psi_{d_s} \\ \psi_{q_s} \end{bmatrix} = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} i_{d_s} \\ i_{q_s} \end{bmatrix} \begin{bmatrix} \psi_f \\ 0 \end{bmatrix} \quad (16)$$

On other hand and as is expressed in (17):

$$[P(\theta_m)][P(\theta_m)]^{-1} \frac{d}{dt} [\psi_{d_s}] = \frac{d}{dt} (\theta_m) \cdot P(\frac{\pi}{2}) [\psi_{d_s}] = \omega_s [\psi_{d_s}]' \quad (17)$$

$$\begin{cases} v_{d_s} = -\omega_s \cdot \psi_{q_s} + (R_s + pL_d) i_{d_s} \\ v_{q_s} = \omega_s \cdot \psi_{d_s} + (R_s + pL_q) i_{q_s} \end{cases} \quad (18)$$

Where:

$$\omega_s = n_p \cdot \omega_m \quad (19)$$

Substituting (16) in (18), we obtain:

$$\begin{cases} v_{d_s} = L_{q_s} i_{q_s} - \omega_s \cdot (R_s + pL_d) i_{d_s} \\ v_{q_s} = \omega_s \cdot L_{d_s} i_{d_s} + (R_s + pL_q) i_{q_s} + E_s \end{cases} \quad (20)$$

The mathematical relationship of electromagnetic torque in the d-q axis is expressed by :

$$T_e = \frac{3}{2} n_p [\psi_d i_{q_s} + (L_d - L_q) i_{d_s} i_{q_s}] \quad (21)$$

The expression of (21) can be reduced to (22) by putting $i_{d_s} = 0$:

$$T_e = \frac{3}{2} n_p \psi_f i_{q_s} \quad (22)$$

2.2.3. Machine side converter (MSC) control strategy

To adjust the PMSG torque, the rotation speed is controlled to follow the maximum power while the wind speed changes [13, 15] and this is done by the MSC control. This strategy is based on a double closed loop control (Current and speed control) [28]. In the analysis presented in the following, we will use the OTSR. From Figure 6, we have $C_t(\lambda)$ is maximal for optimum values of (λ_{opt}) and ω_m^* can be expressed as:

$$\omega_m^* = \frac{\lambda_{opt} \cdot v}{R} \quad (23)$$

According to our system, the current inner loop control for the direct and quadrature components has the same dynamics to utilize the fast response. The PI controller regulates error results [29].

$$\begin{cases} i_{q_s}^* = PI(\omega_m^* - \omega_m) \\ i_{d_s}^* = 0 \end{cases} \quad (24)$$

The two voltages ($v_{d_s}^*$) and ($v_{q_s}^*$) which are described in (20), are calculated as below:

$$\begin{cases} v_{q_s}^* = PI(i_{q_s}^* - i_{q_s}) \\ v_{d_s}^* = PI(i_{d_s}^* - i_{d_s}) \end{cases} \quad (25)$$

To control the MSC, from (20) we have the modulating signal ($u_{abc_s}^*$):

$$u_{abc_s}^* = [P(\theta_m)]^{-1} X u_{dq_s}^* \quad (26)$$

2.2.4. PI regulators synthesis

To simplify controller design, the PI controller is analyzed in the synchronous d-q frame. The torque control strategy adopted in many studies uses two separate current regulators for i_{ds} and i_{qs} [30]. In other hand; after compensation, (16) becomes:

$$\begin{cases} v_{d_s} = L_d \frac{di_{d_s}}{dt} + R_s i_{d_s} \\ v_{q_s} = L_q \frac{di_{q_s}}{dt} + R_s i_{q_s} \end{cases} \quad (27)$$

According to (27), we have [13, 21]:

$$\frac{i_{q_s}}{v_{q_s}} = \frac{1}{sL_q + R_s} \quad (28)$$

$$\frac{i_{d_s}}{v_{d_s}} = \frac{1}{sL_d + R_s} \quad (29)$$

Figure 7 represents the MSC control block diagram where there are two similar PI regulators which controls the direct and quadrature components.

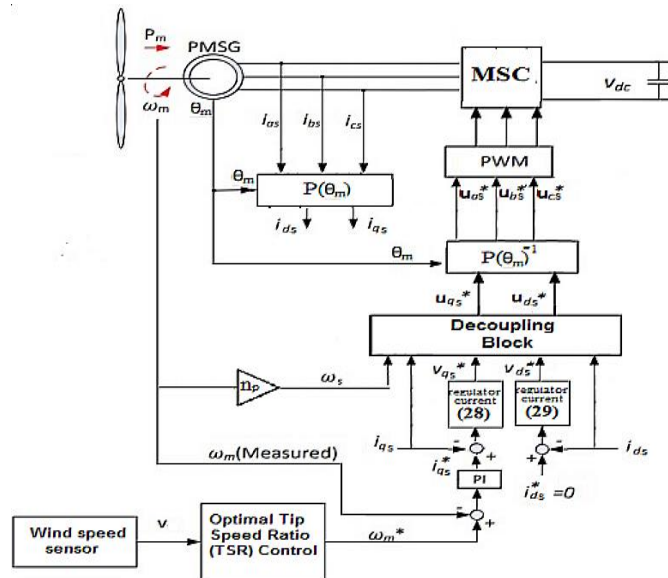


Figure 7. MSC control block diagram

2.2.5. GSC control using resonant controller

The objective of the GSC is to maintain the continuous bus voltage constant and to control the active and reactive power to be injected into the network [31]. In this method, the direct park transform is not used. From Figure 7, we obtain:

$$\begin{cases} i_{qg}^* = PI.(v_{dc}^* - v_{dc}) \\ i_{dg}^* = 0 \end{cases} \quad (30)$$

Applying Inverse Park transform, the grid currents references in 'abc' coordinate, (i_{abcg}^*) are deduced in (31)

$$i_{abcg}^* = [P(\theta_g)]^{-1} X i_{dqg}^* \quad (31)$$

$$v_{abc_i}^* = (i_{abcg}^* - i_{abcg})(k_p + k_i \cdot \frac{2s}{s^2 + \omega^2}) \quad (32)$$

We have to point out, that with this method the problem of the coupling term is not posed.

3. RESULTS AND DISCUSSION

The simulation results under MATLAB/simulink is shown in Figure 8 to Figure 21. The simulation time of the system is set to 3s with an abrupt wind speed profile in order to check the characteristics of the system in order to maintain stable operation.

Figure 11 shows that the PLL, successfully tracking the new network frequency as expected. The accuracy of the PLL in detecting the phase angle of the power grid is therefore demonstrated. The following simulation results are to be demonstrated if the change in frequency does not affect the behavior of the entire system.

The simulation of our hybrid SEH system is proposed; both systems (PV and wind) are connected to the network via the AC bus. In our study, we chose to inject all the active powers resulting from the production of photovoltaic and wind systems by fixing the reactive powers of the two zero systems. In addition, the simulation results of the SEH are virtually the same as for each system operating on its own. This is the strong point of this topology, that is to say in addition to the simplicity of the command, if one system at a given moment will fail, the other remains operational, which is not the case for the DC bus architecture. The Active and Reactive powers delivered by the wind and PV systems, which are injected into the electrical grid, was represented by Figure 15 and Figure 20 respectively. While all the resulting Active and Reactive powers of the PV and wind systems are injected into the electrical grid, was represented by Figure 21.

- From 1.8 to 2s, $G = 1000 \text{ w/m}^2$, $V = 13 \text{ m/s}$, $P_{1\text{ref}} = 9320 \text{ W}$, $P_{2\text{ref}} = 2000 \text{ W}$, $f = 50.2 \text{ Hz}$: The active powers to be delivered by the PV and wind systems are respectively 2000 W and 9320W.
- From 2 to 3 s, $G = 1000 \text{ w/m}^2$, $V = 15 \text{ m/s}$, $P_{1\text{ref}} = 14320 \text{ W}$, $P_{2\text{ref}} = 2000 \text{ W}$, $f = 50 \text{ Hz}$: The active powers to be delivered by the PV and wind systems are respectively 2000 W and 9320W. According to Figure 21, regardless of the evolution of irradiation and wind speed, the reactive power injected into the network remains zero. This result confirms that the decoupled current control method is very precise in the PPM follow-up.

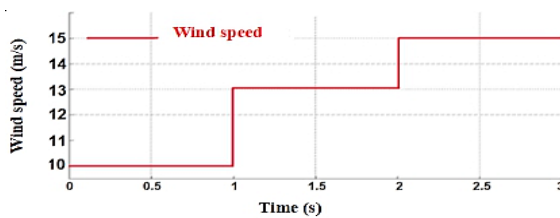


Figure 8. Wind speed profile

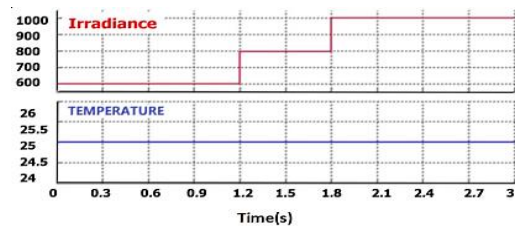


Figure 9. Solar irradiation and temperature profile

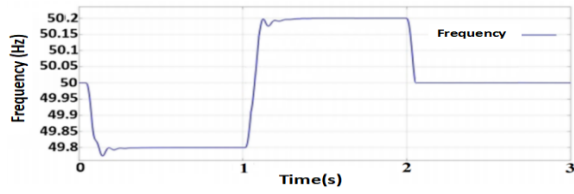


Figure 10. Network frequency detected by PLL

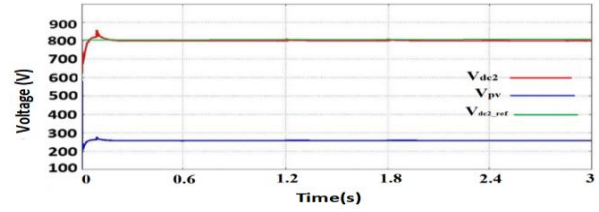


Figure 11. Voltage on DC side

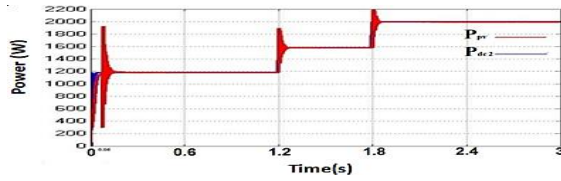


Figure 12. Power on DC side

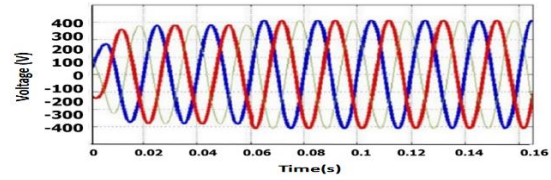
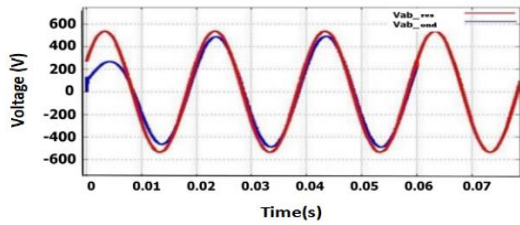
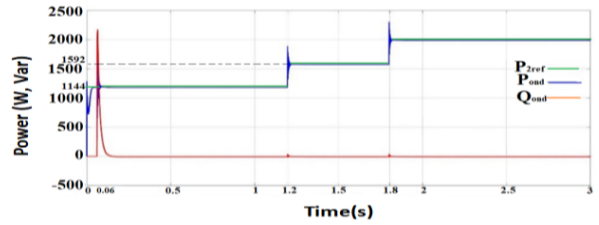
Figure 13. Inverter voltage (V_{abc_ond})Figure 14. Network and inverter voltage (V_{ab_grid}/ V_{ab_ond})

Figure 15. Active and reactive powers delivered by the PV inverter

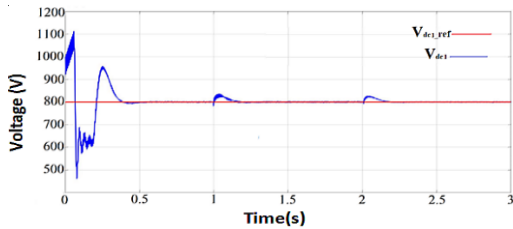


Figure 16. Reference voltage and continuous bus voltage of the wind system

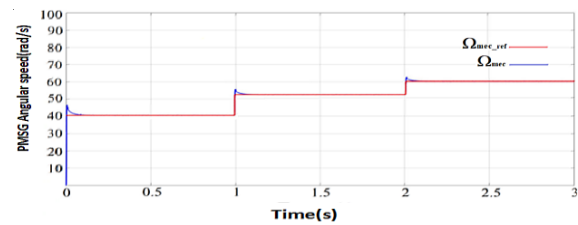
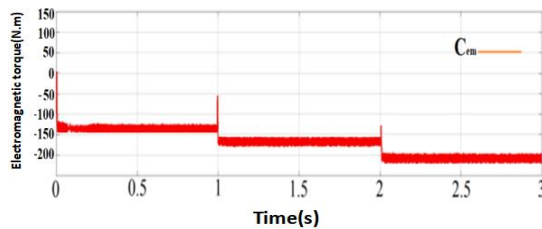
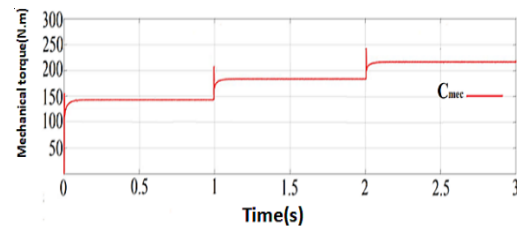


Figure 17. Reference Angular Speed and PMGS Rotor Angular Speed

Figure 18. Electromagnetic torque (C_{em})Figure 19. Mechanical torque (C_{mec})

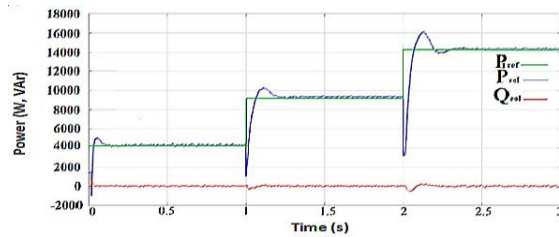


Figure 20. Active and Reactive Powers delivered by the wind system

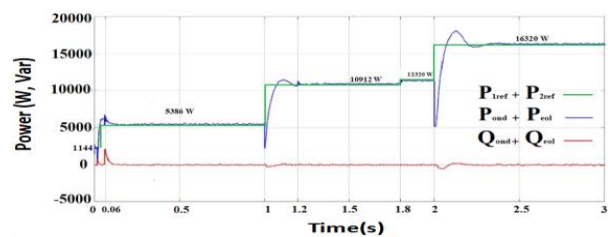


Figure 21. HES Active and Reactive Powers Injected into the Network

4. CONCLUSION

The present work concerns the most current application of renewable energies, that of the simulation and control of SEH connected to the electricity distribution network. Our study was limited to two systems: PV system and Wind system connected to the power grid via the PCC. Control strategies and simulation are presented. The grid-connected photovoltaic system without a transformer, without storage system, using a simple and easy to MPPT controller, can be considered a very interesting topology for such applications.

The simulation results show that the active powers produced by each of the two PV and wind systems are effectively added and injected into the network. The reactive power resulting from the reactive powers of both systems is maintained at zero, despite sudden variations in solar irradiation and wind speed. Although this AC Bus architecture is more expensive than the DC Bus architecture, the simplicity of ordering to be a relevant advantage for our choice. It shows also the correct tracking accuracy and fast response with MPPT technique. The PLL allows to extract the phase angle of the network with preview. By quickly synchronizing the PV system to the network.

The voltage of the continuous bus, which is a critical parameter for integrating such an energy system into the power grid, is kept constant. The results also show that the active output power is stable and that the wind turbine can operate with high efficiency as expected. Finally, the use of a resonant controller for the network-side converter (GSC) is advantageous, given the simplicity of its design and the speed of simulation. The performance of the power control of the SEH with injection of the entire production to the grid is conclusive with a good follow-up of the instructions and rejection of the disturbances, despite the sudden variations in frequency, irradiation and wind.

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